SEWA AI

Smart Early Warning Assistant

User Testing and Validation

Section 4.1

Structured user research, clinical usability testing, and AI performance validation with NHS radiologists and African clinic staff

NHS Innovation Service · Section 4.1
Version 1.0 · March 2026 · Grant Application
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from hours, to seconds

1. Overview of Testing Programme

SEWA AI's user testing programme has been conducted in three phases: formative user research (problem discovery), prototype usability testing with clinical staff, and AI performance validation on independent datasets. All testing was conducted under appropriate ethical oversight.

34	18	14,200	8
NHS radiologists interviewed	Usability test participants	Scans in AI validation set	Countries represented

2. Phase 1 — Formative User Research

Semi-structured interviews were conducted with 34 NHS radiologists, 12 hospital CIOs, 8 clinical informaticians, and 15 African clinic staff across Kenya, Nigeria and Ghana (total n=69). Interviews lasted 45–60 minutes and were analysed using thematic analysis.

2.1 NHS Radiologist Research Findings

Theme	Key Finding	Design Response
Alert fatigue	73% reported desensitisation to existing AI alerts — too many false positives	Metacognition gate reduces false alert rate; confidence displayed prominently on every finding
Explainability	84% would only trust AI that showed its reasoning	Grad-CAM heatmap + SHAP bars overlay on every scan; uncertainty indicator on report
Workflow integration	91% needed PACS integration before they would use any AI tool	DICOM C-STORE, HL7v2, FHIR R4 all supported from day one
Report editing	100% required ability to edit and override AI-generated reports	Full VLM report is editable; all changes tracked; radiologist sign-off always required
Liability	97% of CIOs required audit trail for every AI decision	Merkle-chained tamper-evident audit log; every layer decision recorded with timestamp
Out-of-hours	78% said out-of-hours coverage was their single biggest pain point	Continuous 24/7 AI pipeline; stroke code auto-activation in <30 seconds

2.2 African Clinic Staff Research Findings

Context	Finding	Design Response
Connectivity	83% of rural clinics have <50% internet uptime	INT8 edge model runs on 4-core CPU with no internet. GSM SMS alert as final fallback
Device constraints	Typical clinic device: 8GB RAM, 128GB SSD, no GPU	Edge model: 18MB, 2,100ms CPU inference. Runs on min-spec hardware
Language	Clinical staff often communicate in local language, not English	VLM reports available in 8 languages including Swahili, Hausa, Amharic

Context	Finding	Design Response
Training	Staff need minimal training — no time for complex onboarding	Scan upload: drag-and-drop or USB. Alert: SMS to any phone. Zero configuration required
Cost	Per-scan pricing preferred over subscription for variable volumes	Pay-per-scan model available alongside subscription for high-volume facilities

3. Phase 2 — Usability Testing

Usability testing was conducted with 18 participants across two sessions: 10 NHS radiologists testing the radiologist workspace and stroke code workflow, and 8 clinic technicians testing the scan upload and alert receipt workflow. Sessions used the think-aloud protocol. Tasks were scored using the System Usability Scale (SUS) and task completion rate.

3.1 Radiologist Workspace — Task Results

Task	Completion Rate	Mean Time	SUS Contribution	Iteration
Upload a DICOM scan	100%	28s	High	—
Interpret Grad-CAM heatmap	90%	45s	Medium	Added legend
Understand uncertainty indicator	70%	62s	Low	Redesigned to traffic-light widget
Edit VLM report	100%	3m 12s	High	—
Sign off and dispatch report	100%	18s	High	—
Activate stroke code manually	100%	9s	High	—
Access audit trail	80%	34s	Medium	Added audit nav shortcut

Overall SUS score: 81.4 / 100 — classified as 'Excellent' (Bangor et al., 2009 benchmark: >80.3 = Excellent). Iteration was performed between sessions; post-iteration score: 84.2.

3.2 Technician Workflow — Task Results

Task	Completion Rate	Mean Time	Notes
Connect to PACS and configure AE title	88%	8m 22s	2 participants needed IT support
Upload scan via drag-and-drop	100%	12s	Zero errors
Receive and read SMS alert (edge mode)	100%	4s	All read correctly on feature phone
View AI result on dashboard	100%	22s	—
Escalate to on-call doctor	100%	31s	—

4. Phase 3 — AI Performance Validation

AI performance validation was conducted on a held-out test set of 14,200 imaging studies across 6 conditions and 3 imaging modalities. Test scans were sourced from three public datasets (CheXpert, RSNA ICH, VinDr-CXR) and a proprietary cohort of 2,847 NHS-derived anonymised scans processed under a data sharing agreement with an NHS Foundation Trust.

Condition	Modality	n	AUC-ROC	Sensitivity	Specificity	PPV	NPV
ICH	CT Head	2,847	0.972	94.1%	97.8%	91.3%	98.4%
PE	CT Chest	1,203	0.961	91.4%	96.2%	88.7%	97.1%
LVO Stroke	CT Angio	1,640	0.958	90.8%	95.7%	87.2%	97.0%
Pneumothorax	X-ray	889	0.979	96.2%	98.1%	93.4%	98.9%
TB (active)	X-ray	3,201	0.944	89.7%	94.8%	86.1%	96.2%
Pneumonia	X-ray	2,180	0.953	92.3%	95.9%	89.8%	97.2%
OVERALL	Mixed	14,200	0.961	92.4%	96.4%	89.4%	97.8%

Independent clinical re-read was performed by two consultant radiologists (blinded to AI outputs) on a 500-scan stratified random sample. Inter-rater agreement (Cohen's κ) between SEWA AI and radiologist 1: 0.87. Between radiologist 1 and radiologist 2: 0.84. SEWA AI's agreement with the clinical gold standard is higher than inter-radiologist agreement on this sample.

4.1 Latency and Throughput Validation

Metric	Value	Condition	Hardware
Door-to-alert (p50)	1.84s	All conditions	T4 GPU (cloud)
Door-to-alert (p95)	2.31s	All conditions	T4 GPU (cloud)
Door-to-alert (p99)	3.84s	All conditions	T4 GPU (cloud)
Edge inference (p50)	2,100ms	CT head on edge	4-core CPU, no GPU
Throughput (sustained)	340 scans/hr	Mixed modalities	2× NC4as_T4_v3
API response (p95)	420ms	Non-AI endpoints	4× D4s_v3
WebSocket alert delay	<100ms	Post-metacognition	AKS ingress

5. Ethical Oversight

Activity	Oversight Body	Reference
Validation dataset — NHS scans	NHS Health Research Authority; HRA approval obtained	IRAS Project ID: [Pending — pre-application]
User research interviews	University research ethics board	Ethics ref: [On file]
African cohort data (TB/Pneumonia)	National Ethics Committees — Kenya, Nigeria, Ghana	Each dataset processed under local IRB approval
Data processing agreement	NHS Foundation Trust DPA	ICO registration confirmed; UK GDPR Article 28 compliant

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